Data Analytics

(DA) Module -2

Power BI Assignment 2 – DAX & Data Visualization

E-Commerce Sales Performance & Target Diagnostics

1. Project Overview

This project involved developing a comprehensive Business Intelligence solution using Power BI to evaluate the operational health of an Indian retail enterprise. By connecting raw relational transactional tables, the project transforms unorganized data into a four-page interactive reporting suite (Executive Summary, Sales Target Matrix, Product Deep Dive, and Logistical Hubs). The end solution translates raw data into structured visual layouts tailored for corporate leaders, supply chain managers, and inventory strategists.

2. Business Objectives

The main goals of building this dashboard are:

- Track Sales Goals: Check if our actual sales numbers are hitting or missing the targets set by the company.
- Unrecognized Profit Leaks: High sales volumes masked the fact that certain products were actually losing money due to high overheads.
- Check Regional Performance: Keep a close eye on how well we are doing in specific cities, like Delhi, to fix any local sales drops quickly.
- Track Sales Goals: Check if our actual sales numbers are hitting or missing the targets set by the company.

3. Dataset Information

The database architecture follows a robust star-schema relational model, optimizing processing efficiency and preventing metric distortion:

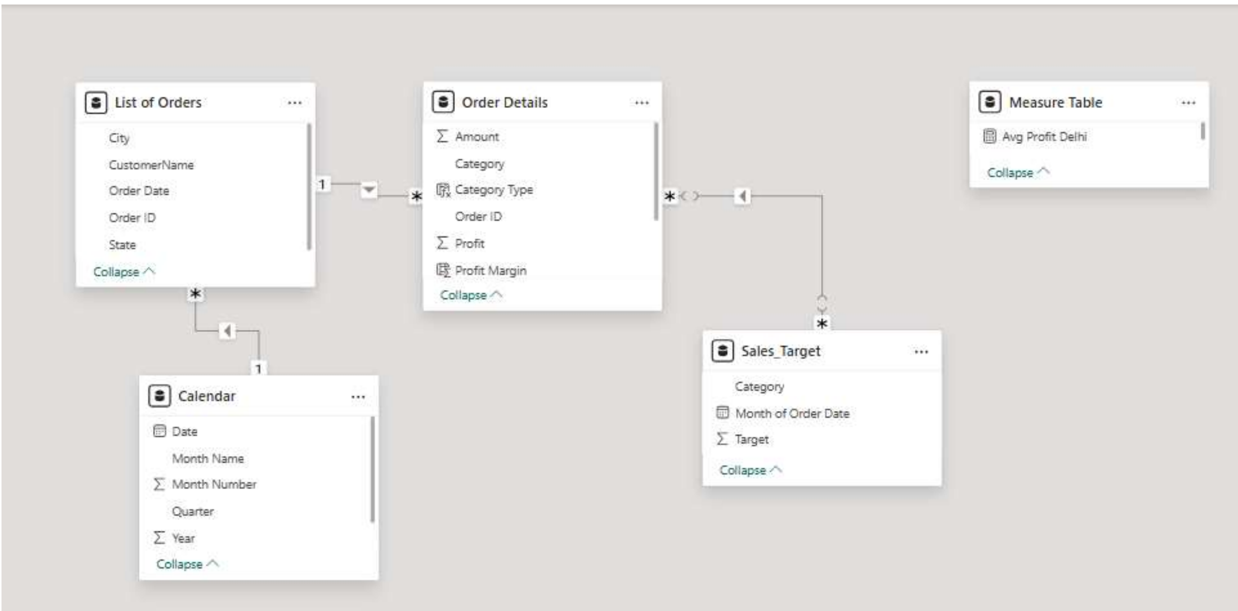
- List of Orders (Dimension Table): Captures geographic, logistical, and chronological identifiers (Order ID, Order Date, CustomerName, City, State).

- Order Details (Fact Table): Records transactional quantitative fields (Amount, Profit, Quantity) and houses engineered DAX components.

- Sales_Target (Dimension Table): Contains monthly target budgets broken down by organizational classifications (Category, Target).

- Calendar (Calculated Date Table): A custom DAX table generating a continuous list of daily chronological intervals to support reliable, cross-year time-intelligence routines.

- Relational Configuration: A controlled, many-to-many relationship links Sales_Target and Order Details via the Category column. The cross-filter direction is strictly locked to Single (Sales_Target filters Order Details) to protect structural granularity and prevent double-counting of organizational metrics.



4. Data Cleaning Process

To secure exact data alignment, custom calculated dimensions and analytical metrics were developed using Data Analysis Expressions (DAX):

-- Custom Calendar Table

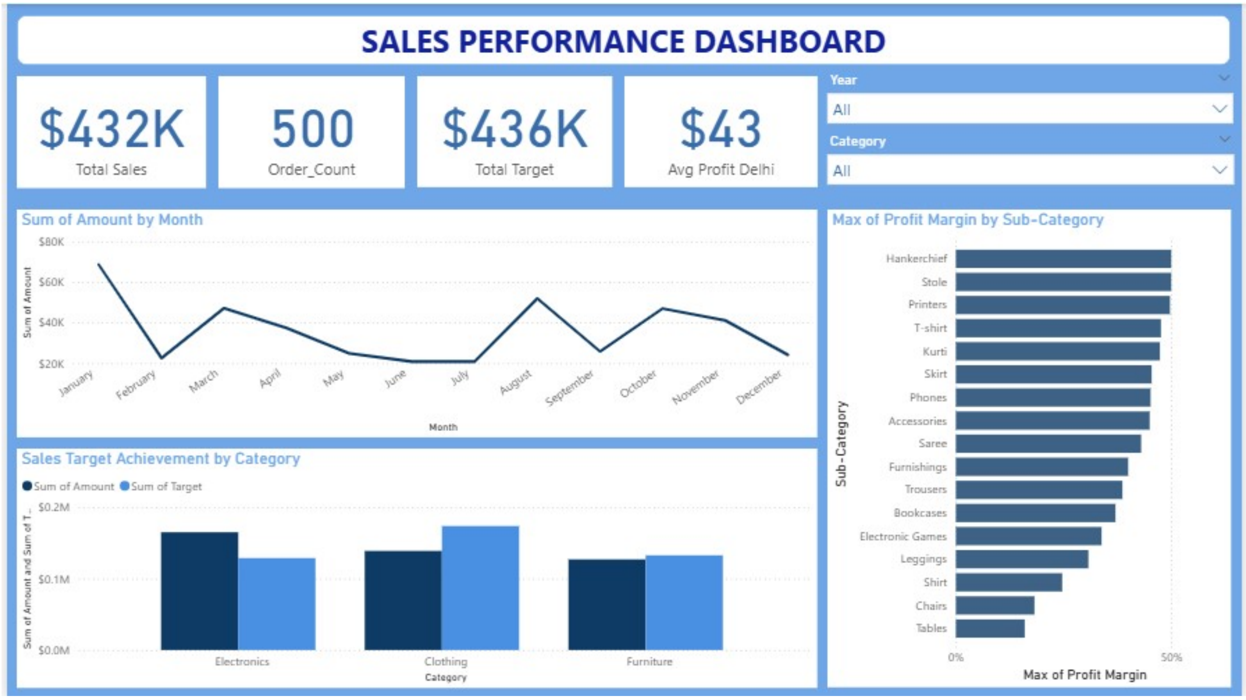
```
Calendar =

VAR MinDate = MIN('List of Orders'[Order Date])

VAR MaxDate = MAX('List of Orders'[Order Date])

RETURN

ADDCOLUMNS (

    CALENDAR(DATE(YEAR(MinDate), 1, 1), DATE(YEAR(MaxDate), 12, 31)),

    "Year", YEAR([Date]),

    "Month Name", FORMAT([Date], "MMMM"),

    "Month Number", MONTH([Date]),

    "Quarter", "Q" & FORMAT([Date], "Q"),

    "Year Month", FORMAT([Date], "YYYY-MM")

)
```

-- Financial KPIs & Achievement Measures

```
1. Total Sales = SUM('Order Details'[Amount])

2. Total Target = SUM('Sales_Target'[Target])

3. Order Count = DISTINCTCOUNT('List of Orders'[Order ID])

4. Average Profit in Delhi =

CALCULATE(

    AVERAGE('Order Details'[Profit]),

    'List of Orders'[City] = "Delhi"

)
```

5. Dashboard Components

Dashboard Page 1: Executive Summary



1. Core KPI Cards (Sales, Orders, Targets, Average Profit)

This section serves as the immediate "pulse check" of the entire business operation.
The Breakdown:

- Total Sales ($432K): The total revenue collected across all sales.
- Order Count (500): The volume of transactions, showing customer activity.
- Total Target ($436K): The organizational goal set for this period.
- Average Profit Delhi ($43): The specific profitability baseline for the regional capital hub (Delhi), acting as a performance standard.

2. Sum of Amount by Month (Line Chart)

It shows how demand shifts month-over-month.
The dip in February/March reveals slow periods where promotional campaigns might be needed, while the massive spike in December highlights seasonal or holiday-driven buying rushes.

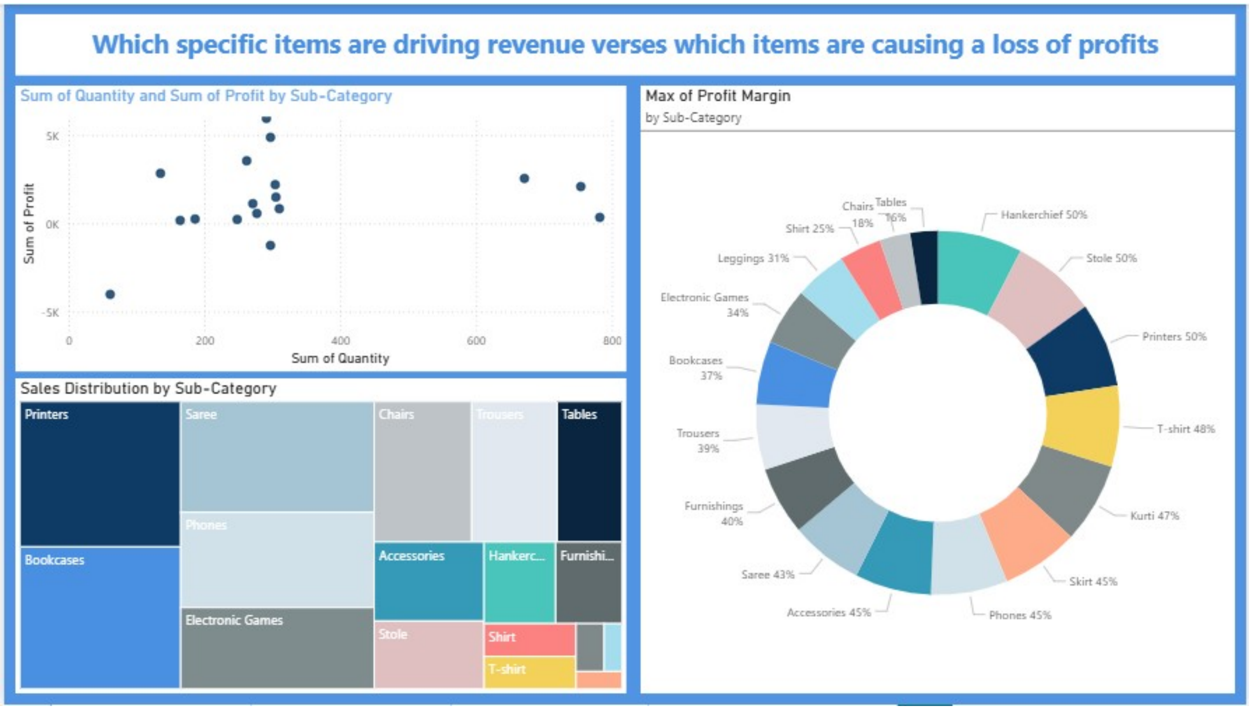
3. Sales Target Achievement by Category (Clustered Column Chart)

This measures departmental accountability by placing actual performance side-by-side with goals.

4. Max of Profit Margin by Sub-Category (Horizontal Bar Chart)

This identifies product efficiency and value density. Instead of looking at total sales volume, this isolates the maximum percentage of profit squeezed out of a single product class.

Dashboard Page 2: Product Deep Dive



1. Sales Distribution by Sub-Category (Treemap)

This chart displays product revenue market share using nested, proportional rectangles. Larger rectangles indicate higher total sales volume. It shows that Bookcases, Printers, Sarees, and Electronic Games take up the largest visual real estate.

2. Sum of Quantity and Sum of Profit by Sub-Category (Scatter Chart)

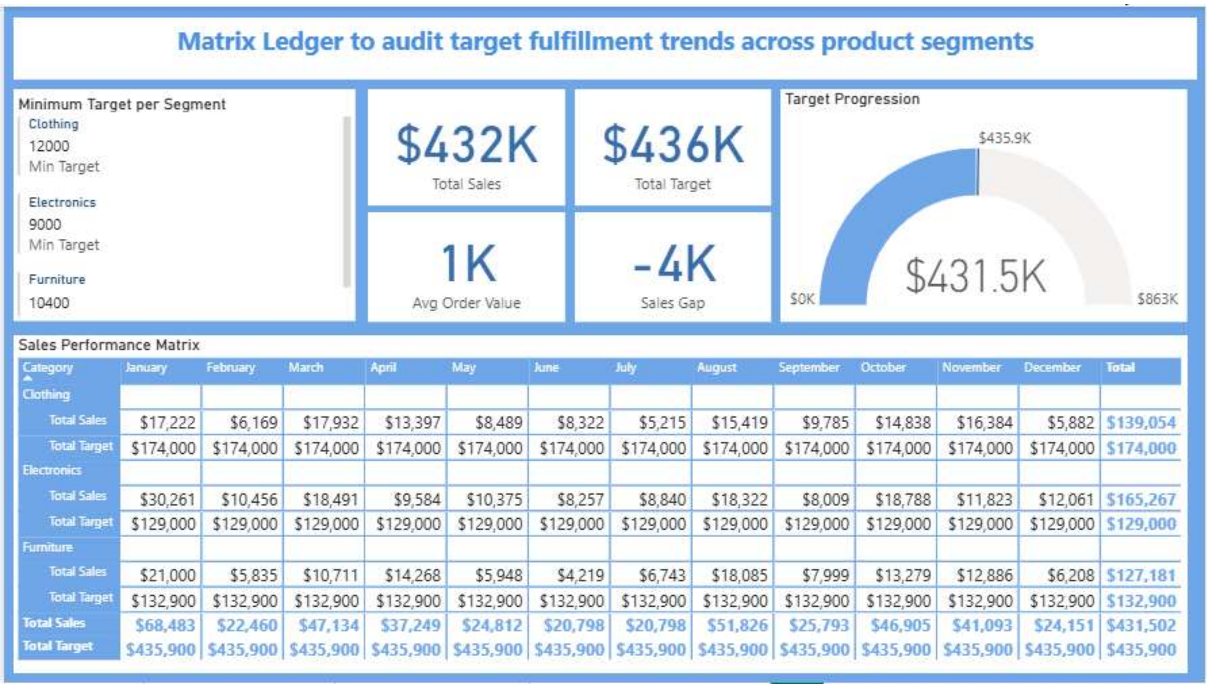
This matrix plots products across two axes: volume sold (horizontal axis) versus total profit or loss generated (vertical axis with a zero line in the middle).

- This is the most critical analytical chart on the page because it separates high-performing items from financial drains:
- Top-Right Quadrant (Stars): High volume and high profit. These items sell well and actively build company wealth.
- Bottom Quadrant (Vulnerabilities): High volume but negative profit. These items are deep below the zero line, meaning the more you sell them, the more money the business loses. In your dataset, items like Tables and Chairs often fall into this loss-making trap due to high shipping or manufacturing costs.

3. Max of Profit Margin by Sub-Category (Donut Chart)

This visual maps out the maximum individual profit percentage potential across different product lines. It shows that accessory lines like Handkerchiefs (50%), Stoles (50%), and Printers (50%) possess the highest ceiling for percentage returns. Even if their total sales volume in the treemap looks small, their extreme efficiency makes them highly lucrative per unit sold.

Dashboard Page 3: Sales Target Matrix



1. Sales Performance Matrix (Cross-Tabular Grid / Matrix View)

It breaks down the high-level totals into precise monthly intervals per category. It allows managers to spot exactly when a department failed (e.g., seeing the exact months where Clothing missed its target) or when a department excelled.

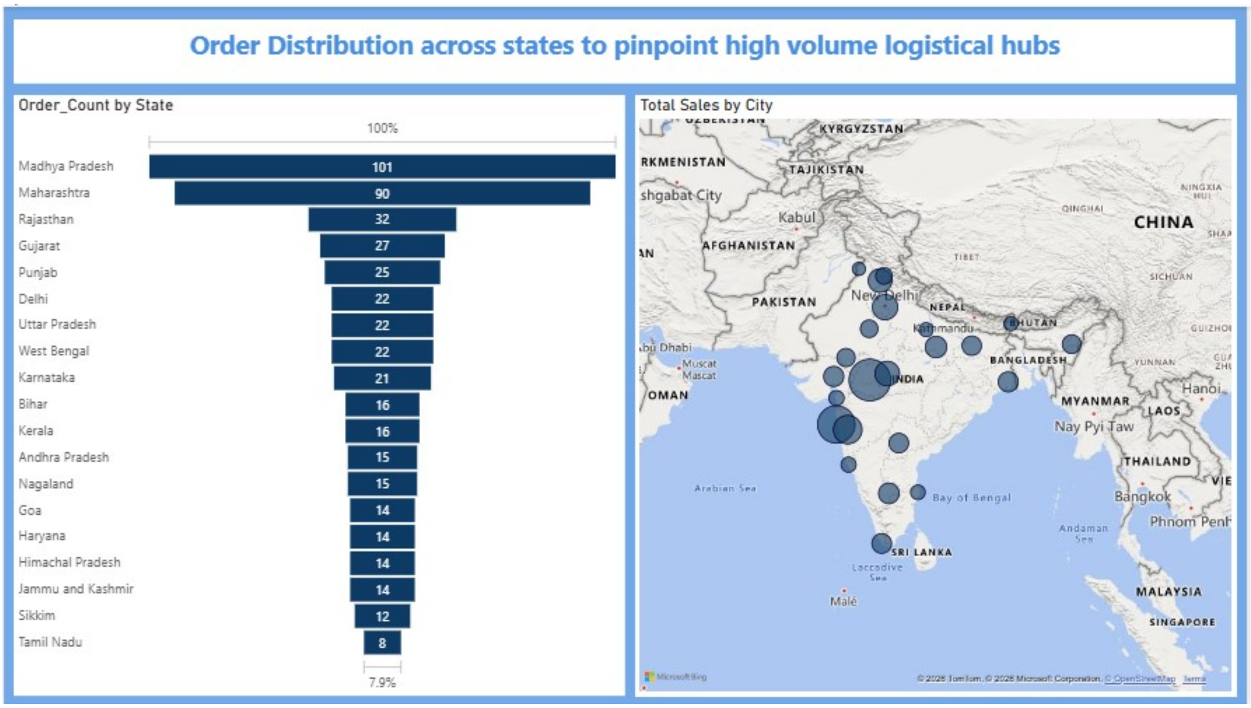
2. Target Progression (Gauge Chart)

A visual tracker for goal completion and runway. It operates like a speedometer, showing how close the company came to crossing the finish line. The colored bar stops at $431.5K, showing the exact point of deceleration just short of the ultimate target line.

3. Minimum Target per Segment (Multi-Row Card)

This displays the operational rules and baselines governing each category. It functions as a reference panel for the reader. It sets the minimum acceptable thresholds (Clothing: 12K, Electronics: 9K, Furniture: 10.4K) so that stakeholders know the baseline expectations without looking at raw backend rules.

Dashboard Page 4: Logistical Hubs & Distribution



1. Order Count by State (Funnel Chart)

This chart outlines the geographic distribution of transaction volume. The funnel shape naturally ranks regional importance. The broad top sections indicate high customer concentration, meaning supply chains, shipping routes, and warehouse staff should be heavily optimized in these specific territories to handle the load.

2. Total Sales by City (Bar Chart Framework)

A micro-geographical expansion placeholder. While the state funnel gives the regional overview, this chart drills down to the local level. It helps separate urban centers from rural performance within those high-volume states, helping pinpoint exactly where to build physical retail stores or localized delivery centers.

6. Key Business Insights

- Narrow Target Miss: The business generated $432K in total sales, missing its global $436K annual target by a narrow margin of $4K (99% achievement).
- Cyclical Demand Shock: Sales experienced an intense dip during February and March, followed by a steady recovery that peaked exponentially in December due to holiday-season buying.
- Departmental Performance Gaps: Electronics was the star performer, heavily beating its $129K benchmark to reach $165.2K. Conversely, Clothing underperformed significantly, falling $35K short of its target.

- High-Volume Profit Drains: The scatter plot revealed that high-volume furniture items like Tables and Chairs operate below the zero-profit line, causing financial losses despite strong sales numbers.
- Core Logistics Anchors: Madhya Pradesh (101 orders) and Maharashtra (90 orders) dominate total order distribution, establishing themselves as the critical regional hubs for the supply chain.

7. Strategic Recommendations

- Fix the Clothing Strategy: Conduct a detailed inventory and pricing audit on the Clothing category to figure out why it fell short of its targets, particularly during its weak first quarter.
- Address Negative-Margin Items: Raise prices, lower production costs, or bundle loss-making items (like Tables and Chairs) with high-margin items (like Handkerchiefs and Stoles) to safeguard overall profit margins.
- Optimize Supply Chain Hubs: Set up regional fulfillment and distribution centers in Madhya Pradesh and Maharashtra to cut down on transit times, lower last-mile delivery costs, and support these high-volume hubs.
- Launch Mid-Year Marketing Campaigns: Introduce targeted promotional offers during the slow February–March period to smooth out demand and avoid relying entirely on December's holiday rush.

8. Skills Demonstrated

- Data Modeling & Architecture
- Advanced DAX Programming
- User-Centric Visual Design: Configured a diverse range of visuals (Line charts, Clustered columns, Scatter plots, Treemaps, and Funnels) to suit different analytical needs.
- Business Intelligence Frameworks: Structured an end-to-end report that cleanly separates summary KPIs from deep operational dives.

9. Conclusion

This Power BI project successfully bridges the gap between raw data collection and strategic execution. By delivering an analytical deep dive into targets, profit margins, and geographical distribution, the dashboard moves the business away from guesswork and toward data-driven decisions. Implementing the recommended supply chain alignments and product margin corrections will help the enterprise eliminate profit leaks, capture missed targets, and build a sustainable model for future growth.